

Lab 08: Coding for Synthesis

Assigned on Oct 23, 2025

Due day in **Oct 30, 2025**

Objective

In this lab, you will learn:

1. Implementation and analysis different hardware architecture of digital filter.

Demo checklist:

- ☐ Show simulation results and performance of the one multiplier and one adder design.
- ☐ Show simulation results and performance of the two-parallel architecture.
- ☐ Answer the following questions
 1. Record different hardware architectures (direct, one multiplier and one adder, pipeline, parallel).
 2. Analyze the advantage or disadvantage of each hardware architecture.
 3. Check the *resources report* in synthesis report, analyze the adder/multiplier which are used in both of your designs. Is there any resource sharing in your design?

Environment Setup

Copy lab file packages from ee4292. Decompress the package and enter it. You can check the file list in Appendix.

```
$ cp ~ee4292/iclab2025/lab08.zip .
```

```
$ unzip lab08.zip
```

```
$ cd lab08/
```

Note: please run simulation in the *sim* directory to maintain a fine data management.

Description

In lab6, we have implemented the filter for ECG application. And we are now going to further explore the design space of the filter architecture. As shown in the optional action item in lab8, there are many possible architectures to improve the performance of the filter. You are going to implement a **one multiplier and one adder architecture** and a **two-parallel architecture** of the filter in this lab. The corresponding architecture diagram are shown in Fig 1 and 3. You can start from the fir1.v which is your finished

version of lab6. Since the output behavior of parallel architecture are different from the last lab, you may need to modify the finite state machine in this lab.

Action Items

I. One multiplier and one adder architecture.

- Please use only one multiplier and one adder architecture as shown in figure 1. You can refer to figure 2 to check your function correctness. (Note: If you need longer simulation time, please modify the “END_CYCLE” in fir_test.v)

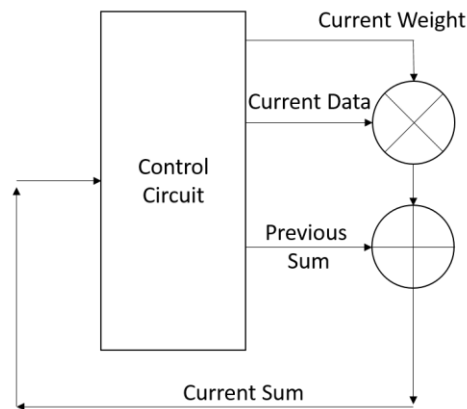


Figure 1. one multiplier and one adder architecture.

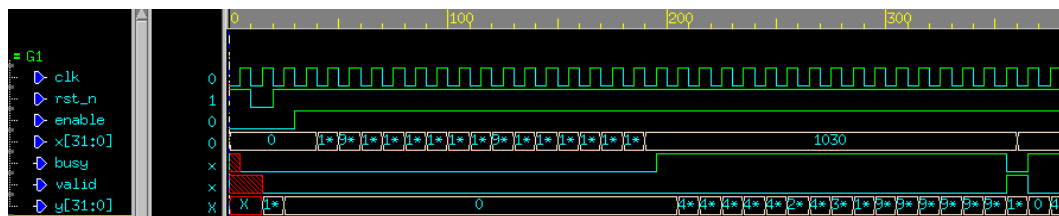


Figure 2. Waveform of one multiplier and one adder architecture.

- Record your synthesis result.

Area (μm^2)	Timing (ns)	Cycles (#)	Performance

II. Parallel architecture.

- Parallel structure is a MIMO system, multiple inputs can be processed in one cycle, as a result, the overall throughput is improved. Figure 3 is the parallel architecture, which can be expressed as follows:

$$y[2n] = \sum_{i=0}^{15} x[2n - i] * a_i$$

$$y[2n+1] = \sum_{i=0}^{15} x[2n-i+1] * a_i$$

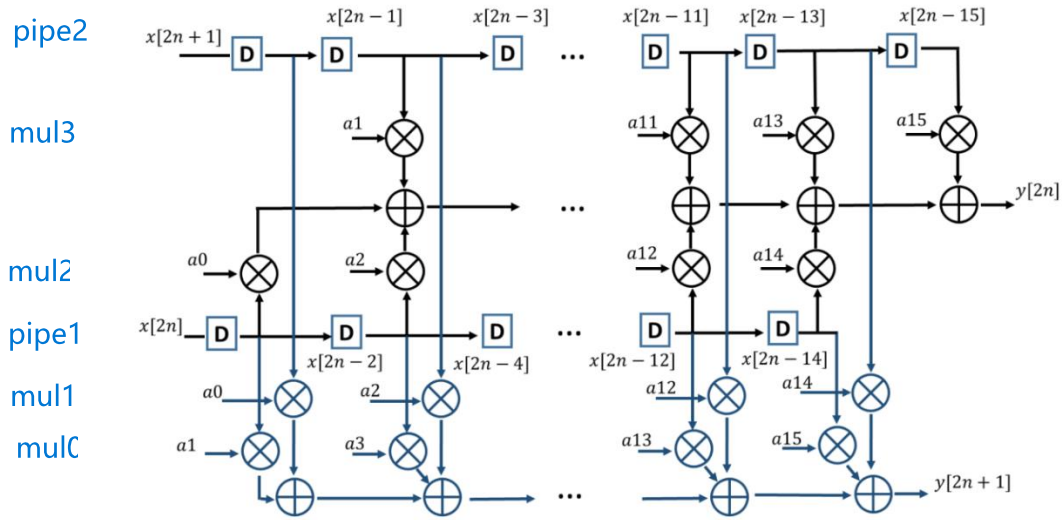


Figure 3. Parallel structure.

- The circuit I/O should be modified to feed and receive multiple data in one cycle.

I/O name	Number of bits	Description
clk	1	Clock input
rst_n	1	Active low reset
enable	1	Module enable
x0	N	ECG data input
x1	N	ECG data input
busy	1	Module busy
valid	1	Output data valid
y0	N	Processed data output
y1	N	Processed data output

- Figure 4 is the waveform of parallel architecture. You can refer to it to check your function correctness.

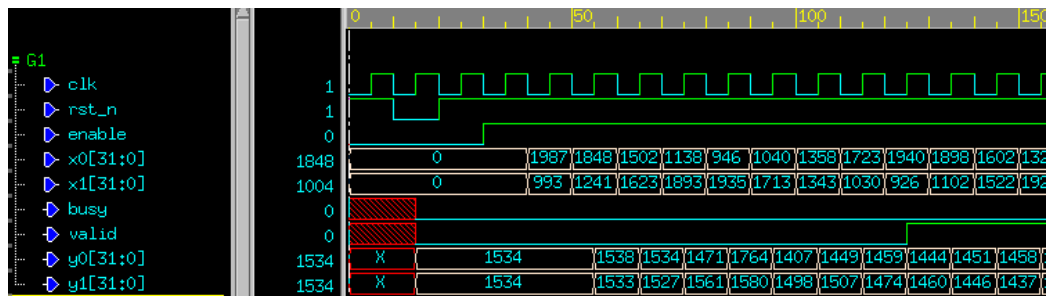


Figure 4. Waveform of the parallel architecture.

4. The parallel architecture is obviously a trade-off between power and area, so apart from comparing the performance index, try to synthesis the parallel architecture with double-period clock, and see how much power is saved (with the same throughput as the original architecture). Record your synthesis result.

Area (μm^2)	Timing (ns)	Cycles (#)	Performance

Appendix

Directory	Filename	Description
hdl	fir1.v	Your design here
hdl	fir1_parallel.v	Your design here
sim	fir1_parallel_test.v	Testbench
sim	fir1_test.v	Testbench
sim	sim.f	File list and Testbench
sim	sim_parallel.f	File list and Testbench
sim	ecg.csv	Test pattern
sim	data_truth.csv	Golden response
syn	0_readfile.tcl	Synthesis scripts
syn	1_setting.tcl	Synthesis scripts
syn	2_compile.tcl	Synthesis scripts
syn	3_report.tcl	Synthesis scripts
syn	synthesis.tcl	Synthesis scripts
syn	run_dc.bat	Synthesis command
syn	.synopsys_dc.setup	Synthesis setup file